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Enhanced Production of Fatty Acid Methyl Esters And Biochar via Pyrolysis of Palm oil Biomass: an Experimental Analysis

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Abstract The pyrolysis of palm waste offers a sustainable alternative to the open burning of over 1.1 million tons of biomass annually, while simultaneously producing biochar and value-added chemicals. Among these products, fatty acid methyl esters (FAMEs) are particularly attractive as biodiesel precursors derived from the thermal conversion of lipid-containing biomass. In this study, the influence of key operational factors—pyrolysis temperature, feeding rate, and nitrogen atmosphere—was systematically evaluated to optimize FAME and biochar yields. The pyrolytic products were characterized using Thermogravimetric Analysis (TGA), Field Emission Scanning Electron Microscopy (FESEM), and Gas Chromatography–Mass Spectrometry (GC–MS). Thermal analysis showed that the presence of nitrogen enhanced FAME production and improved the thermal stability of biochar, while increasing the temperature decreased FAME formation but promoted aromatic compounds. Morphological observations revealed the development of more crystalline, porous biochar structures at higher temperatures under nitrogen flow. GC–MS analysis indicated that the highest yields of FAMEs and functional biochar were achieved at 450 °C, likely via in-situ methylation of triglycerides facilitated by lignin methoxy groups.

Keywords Pyrolysis · Fatty acid methyl ester · Biochar · Palm oil residue

Abbreviations

ASTM	American Society for Testing and Materials
CHP	Combined Heat and Power
DTG	Derivative Thermo Gravimetric
EDX	Energy Dispersive X-Ray
FAME	Fatty Acid Methyl Ester
FESEM	Field emission scanning electron microscopy
FTIR	Fourier Transform Infrared Spectroscopy
GC-MS	Gas Chromatography–Mass Spectrometry
OFAT	One Factor at a Time
PAHs	Polycyclic Aromatic Hydrocarbons
PKS	Palm kernel shell
VOCs	Volatile Organic Compounds
SEM	Scanning Electron Microscope
TGA	Thermo Gravimetric Analysis

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1 Introduction

Burning agricultural waste has raised environmental and sustainability concerns due to its potential impact on ecosystems, particularly water and air quality. The annual production of over 40 million tons of agricultural plants in Iran contributes to approximately 290 gigagrams of carbon dioxide emissions into the atmosphere. More than one million tons of palm date waste are generated annually in southern Iran, representing an important biomass resource (Abbasi et al., 2019). Consequently, waste management has emerged as one of the most serious environmental challenges. Global initiatives toward greener economies and sustainable development are being reinforced by the growing demand for clean and renewable energy (Onokwai et al., 2023). Thermal conversion technologies, including pyrolysis, have been increasingly utilized to address climate change, enhance energy security, and promote circular bio-economy practices. Biochar and bio-oil derived from biomass pyrolysis are sustainable by-products that offer solutions to both domestic and global waste management issues. Recent studies have emphasized pyrolysis as a promising approach to valorize agricultural waste while minimizing greenhouse gas emissions and costs (Rajak et al., 2025). Beyond its contribution to waste valorization and carbon sequestration, biochar has shown remarkable potential in environmental remediation (Hosseinahli et al., 2021). Biomass-derived biochar possesses a large surface area, porous structure, and reactive functional groups that enable efficient adsorption of heavy metal ions, organic contaminants, and pesticides from wastewater and soils. Such properties make biochar a sustainable material for pollution control, resource recovery, and soil restoration (Muhammed et al., 2025). Moreover, biochar is widely available, and its favorable chemical and physical properties allow it to effectively immobilize heavy metals, remove organic pollutants and pesticides, and contribute to carbon sequestration and greenhouse gas mitigation. Its effectiveness depends on production conditions: high-temperature biochar enhances surface area, hydrophobicity, and microporosity for organic pollutant adsorption, whereas low-temperature biochar is more efficient for inorganic pollutant removal through precipitation, electrostatic interactions, and oxygenated functional groups (Farooq et al., 2025).

Thermal conversion technologies are introduced to address climate change, energy security, and circular economy products. Biochar and bio-oil from the pyrolysis process are valuable waste products and offer a solution to domestic and international waste management issues. Today, pyrolysis technology is a promising approach to the circular bio-economy, which can reduce costs and emissions from agricultural wastes. Additionally, burning agricultural waste can release harmful pollutants such as CO, CH_4, N_2O, CO_2, unburned hydrocarbons, SO_2, NO_X, suspended particles, and, in some cases, PAHs and VOCs into the atmosphere. These pollutants can cause air pollution, respiratory diseases, and discomfort for humans (Rajak et al., 2025; Zhu et al., 2024).

The optimal harnessing of renewable energy sources is crucial for transitioning from fossil fuels to clean energy. Also, Biomass possesses qualities such as biodegradability, affordability, and carbon neutrality, making it a highly promising aspect of biofuels. Unlike other renewable energy sources, bioenergy can be gathered, stored, and transported, enabling its dispatchability and resemblance to conventional fossil fuels (Guo et al., 2023). For instance, a sizeable portion of agricultural waste is burned or used as livestock feed in Jordan during the biomass harvesting process (Al-Bawwat et al., 2023; Al-Hamamre et al., 2017). North America and Iran face significant air pollution from agricultural waste, with CO_2 being the most emitted gas. The high fuel-to-feed ratio of the pyrolysis process makes this technique suitable for generating gasoline or diesel-like fuels (Jahirul et al., 2012). Recently, most efforts have been carried out to achieve high-quality bio-oil and functional biochar from pyrolysis techniques (Chen et al., 2019; Ghorbannezhad & Abbasi, 2021). Various biomass materials undergo pyrolysis to produce biochar with diverse properties that can improve pollutant removal. This method, instead of incineration, helps in carbon sequestration, thereby aiding climate change mitigation, promoting environmental sustainability, and fostering a circular economy (Amalina et al., 2023). Pyrolysis technology enables the conversion of various biomass sources into valuable energy carriers such as bio-oil and biochar. Recent investigations have highlighted the use of hydrochar and biochar-based catalysts in biodiesel production due to their high acidity, surface area, and catalytic efficiency (Aliyu et al., 2025; Yang et al., 2024).

The latest biochar production method utilizes a pyrolysis oxygenation system to generate carboxylic acids and phenolic compounds from lignin. This process enhances the surface area and simplifies pollution disposal. Biochar is also used as a catalytic material for biomass conversion reactions and as an industrial solvent for polyvinyl chloride production (https://www.verifiedmarketresearch.com/product/tetrahydrofuran-market/, 2019; Kim et al., 2019).

Pyrolysis also allows for the production of bio-oil, which is used in industries like transportation, power plants, and refineries (Bharath et al., 2020). Bio-oil is a crucial raw material for CHP power plants, jet fuel, and medical creams. Its oxygenation system enhances the functionality of biochar and its high specific surface area, resulting in useful phenolic compounds and carboxylic acids. Plenty of biomass has been taken into consideration for bio-oil production, such as wheat straw (Dai et al., 2019), corn straw (Liu et al., 2009), rice husk (Bakar & Titiloye, 2013), bagasse (Asadullah et al., 2007), coconut shells (Fardyanti et al., 2018), and waste fish oil (Wisniewski et al., 2010). Although the hemicellulose, cellulose, and lignin content are the main chemical components of biomass, previous research has shown that they can produce valuable products via pyrolysis into bio-oil and biochar (Ghorbannezhad et al., 2023; Onokwai et al., 2023), while the lipid content of palm date residues enables the production of fatty acid methyl esters (FAMEs) by the esterification of fatty acids. Biodiesel is generated by the esterification of lipids into FAMEs. Minimum greenhouse gas emissions, biodegradability, and compatibility with existing motor engines are the main advantages of biodiesel utilization. Understanding pyrolysis parameter conditions is vital for forming FAMEs among different complex compositions containing various chemical substances (Inam et al., 2019). The focus of the current study was on the kinetics of pyrolysis and thermal conversion of date palm residues, which included seeds, leaves, and leaf stems, into valuable products as well as FAMEs. Optimal production conditions based on the effective factors were obtained, and the feasibility and potential of using the products in the production of FAMEs and functional biochar were evaluated and discussed in detail.

In this research, the physical, chemical, and thermal characteristics of palm tree waste were also identified and categorized as approximate, final, and thermal. The parameters affecting the production of bio-oil were assessed using a fluidized bed reactor. The optimization of pyrolysis parameters aims to assess the methylation of fatty acids from lipid-containing biomass as feedstock.

2 Materials and Methods

2.1 Processing and Examination of Biomass

Palm tree waste—including cores, leaves, stems, and other by-products—was collected from several farms in Sistan-Baluchestan province, Iran. This region, located in southeastern Iran, is a major hub for tropical fruit cultivation, especially date palms, resulting in a significant amount of biomass waste. To reduce moisture content, the raw materials were sun-dried for two days and then further dried in a laboratory oven (Memmert UN55, Germany) at 105 °C for two hours, until their moisture content dropped below 10% of the dry weight.

For particle size analysis and subsequent experiments, part of the dried biomass was ground using a high-speed grinder (IKA MF10, Germany) and sieved to achieve a particle size smaller than 0.25 mm. The biomass samples were then subjected to fast pyrolysis under various feeding rates and at temperatures ranging from 450 °C to 550 °C using a fixed-bed pyrolysis reactor (custom-built pilot system, equipped with a temperature control module and nitrogen gas inlet, Iran). Experiments were conducted under both inert (with nitrogen gas) and non-inert (without nitrogen) conditions. Nitrogen gas (purity 99.99%) was supplied by Roham Gas Co., Iran, at a constant flow rate of 200 mL/min to maintain an oxygen-free atmosphere when required.

Experimental design and statistical analysis were performed using Design-Expert software (Version 13, Stat-Ease Inc., USA) to assess the influence of operating variables. Fast pyrolysis was selected due to its operational and economic advantages over other thermochemical conversion methods, including ease of bio-oil storage and transportation. Key design considerations—such as feed drying, particle size, pre-processing, reactor configuration, heat input, residence time, secondary cracking, and separation of biochar and ash—were carefully addressed throughout the process.

Given the abundance and renewability of biomass resources, especially date palm waste in southern Iran, energy generation through biomass conversion presents both environmental and economic promise.

2.2 The Mechanism of Preparation and Pre-Processing of Palm Tree Waste

The pyrolysis process used in this study closely follows the conventional fast pyrolysis procedure, which involves thermal decomposition of biomass in an oxygen-free, sealed reactor system. In the context of simultaneous pyrolysis, three main stages are essential for fuel production: sample preparation, thermal conversion, and product condensation. Prior to pyrolysis, the biomass samples were oven-dried at 105 °C for 24 h using a laboratory oven (Memmert UN55, Germany) and ground to a fine powder using a high-speed grinder (IKA MF10, Germany). In industrial applications, internal heat sources often supply the required energy for drying.

An inert gas, typically nitrogen, was used in this setup to facilitate the rapid transfer of pyrolytic vapors from the high-temperature reaction zone to the condenser. This minimizes secondary cracking reactions and enhances the bio-oil yield. The hot vapor residence time was maintained under 2 s. The nitrogen gas (purity 99.99%; supplied by Roham Gas Co., Iran) was injected at a controlled flow rate of 200 mL/min using a gas flow controller (Cole-Parmer, USA). Flow rate optimization was essential, as excessively high rates can reduce oil yields. The requirement for inert gas varies with the reactor design. For instance, fluidized-bed and circulating fluidized-bed reactors require high gas velocities, while vacuum reactors often operate without inert gases. As noted by Bridgwater et al. (Bridgwater, 2012). Some systems, like spent reactors, may not require nitrogen purging under industrial conditions, but are used in lab-scale setups to control residence time and enhance safety. Feed Rate Control and Safety Protocol: In this study, the biomass feed rate was controlled using a custom-built screw feeder. The feeder's motor speed (in rotations per minute, rpm) was calibrated to the mass feed rate (g/min) by measuring the mass of biomass discharged over timed intervals at various motor settings. The feed rates reported in this study (20–30 g/min) are the calibrated mass flow rates. For each run, 30 g of dried biomass were fed into the pyrolysis reactor. The feedstock was physically isolated from the hot zone by two plungers. It is important to note that the plungers used in the initial experimental setup were fabricated from a compressed asbestos-containing material, selected for its historical use in high-temperature applications. Throughout all experiments, strict safety protocols were enforced to prevent exposure, including the use of a local exhaust ventilation system and appropriate personal protective equipment (e.g., P2 respirators) during handling. Furthermore, following the completion of this experimental campaign and in line with modern safety standards, the asbestos-containing plungers have been permanently decommissioned, safely disposed of, and replaced with plungers made from a certified, non-asbestos, high-temperature ceramic composite material for all future work. To prevent clogging at the end of the feeder, water was sprayed in a circular motion as a cooling barrier.

The reactor was heated by an electrical heating unit with programmable temperature control (Heidolph MR Hei-Standard, Germany), ensuring a consistent heating rate and uniform temperature distribution. The pyrolysis vapors passed through a condenser and heat exchanger system (Liebig-type condenser, Borosil, India), where gases were cooled and condensed into bio-oil and non-condensable gases. A schematic diagram of the pyrolysis setup used in this investigation is presented in Fig. 1. Subsequently, the solid product, pyrolytic biochar, was collected and weighed to assess the yield distribution.

2.3 Study Design

The experimental design was developed to systematically evaluate the effects of pyrolysis temperature and biomass feeding rate on the distribution of major products—bio-oil, biochar, and biogas—derived from palm oil biomass. The purpose of this study was to identify the operating conditions that maximize the production of value-added compounds, particularly fatty acid methyl esters (FAMEs), while ensuring efficient energy recovery and sustainable waste utilization. To achieve this, a response surface methodology (RSM) framework was applied using the data generated from controlled pyrolysis runs. This approach enables statistical modeling of process parameters and their interactions, thereby providing a robust basis for

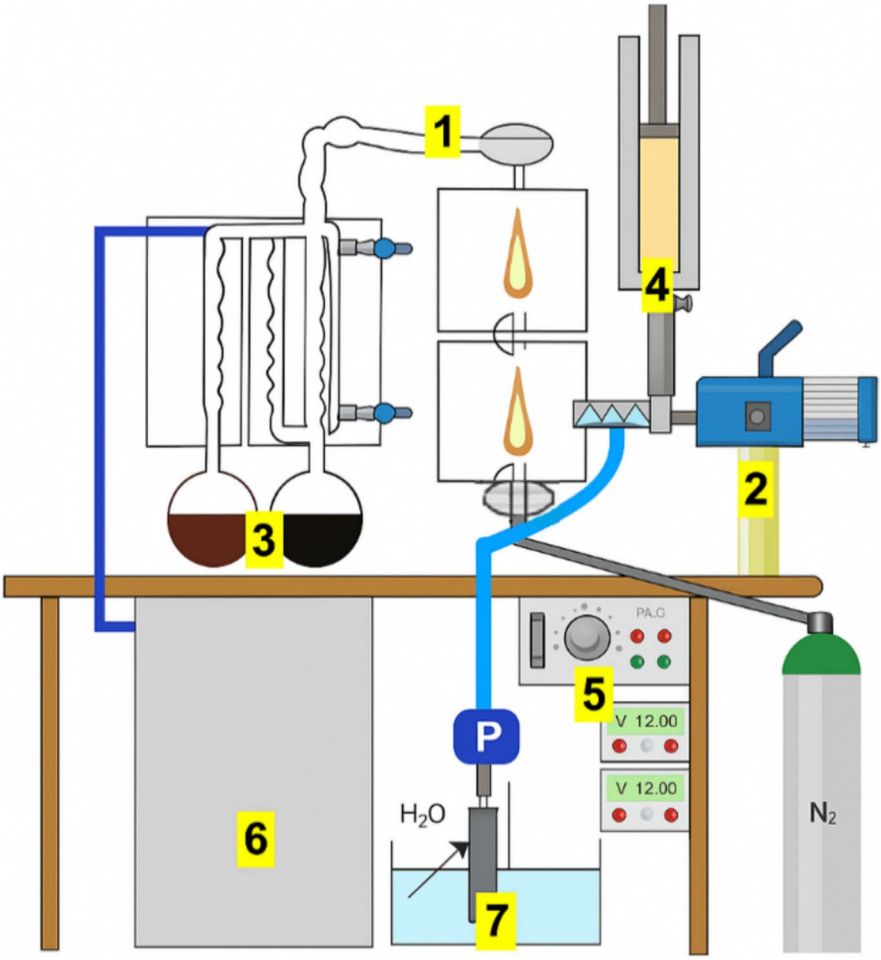


Fig. 1 Schematic diagram of semi-industrial fluidized bed reactor setup: (**4**) Feed tank, (**2**) Motor, (**1**) Catalyst bed, (**3**) Condenser, (**8**) Nitrogen cylinder, (**5**) Temperature controller, (**7**) Water bath, (**6**) Chiller

process optimization. The selection of temperature (450–550 °C) and feeding rate (20–30 g/min) ranges was based on preliminary experiments and literature reports, ensuring coverage of the most relevant operational window for fast pyrolysis of lignocellulosic feedstocks.

Table 1 summarizes the experimental results used for constructing RSM plots. The results indicate clear trends:

- **Bio-oil yield** peaked at 7.10 g under 550 °C and 20 g/min (Run 6), suggesting favorable conditions for liquid fuel precursors.
- **Biogas yield** reached its maximum (12.00 g) at 550 °C and 30 g/min (Run 7), confirming that elevated temperatures and higher feed rates promote cracking of volatiles and secondary gas-phase reactions.
- **Biochar yield** generally decreased with increasing temperature, consistent with enhanced devolatilization and thermal decomposition.

Interestingly, intermediate conditions (e.g., 500 °C and 25 g/min, Run 3) provided a more balanced distribution of bio-oil, biochar, and biogas. This balance is particularly relevant for integrated applications such as combined heat and power (CHP) systems, where co-production of gaseous and liquid fuels alongside functional biochar is desirable.

Overall, the study design provides a structured framework for linking process variables with product yields, enabling both mechanistic understanding and practical optimization of palm biomass pyrolysis for bioenergy and environmental applications.

The data presented in Table 1 were used to construct RSM plots, which allowed the investigation

Table 1 Data are representative values derived from experimental runs used for RSM surface plotting

Run	Temperature (°C)	Feeding rate (g/min)	Bio-oil yield (g)	Biochar yield (g)	Biogas yield (g)
1	450	20	6.50	13.80	7.80
2	450	30	6.20	12.30	8.50
3	500	25	6.80	11.50	9.00
4	510	28	6.95	10.50	10.20
5	530	22	7.05	11.00	10.50
6	550	20	7.10	12.20	10.90
7	550	30	6.60	9.50	12.00
8	500	30	6.70	10.80	11.00
9	470	25	6.40	12.80	9.50

of both individual and interactive effects of pyrolysis temperature and biomass feeding rate on product yields. This analysis provides a clear framework for identifying optimal operating conditions for enhanced FAME production and functional biochar recovery.

Detailed statistical outputs of the RSM modeling, including ANOVA results for the 2FI model (Table S1), model fit statistics (Table S2), and the regression metrics (Table S3), are provided in the supplementary information for more clarification.

3 Results and Discussion

3.1 Evaluation of Pyrolysis Pilot

Pyrolysis is a versatile technology used for producing biofuels and value-added products across industries such as transportation, power generation, and pharmaceuticals. In regions like Iran, where significant amounts of palm waste are generated annually, the thermal conversion of this waste offers a sustainable alternative to conventional disposal methods, reducing environmental burdens while providing economic benefits (Ghorbannezhad et al., 2020). Additionally, pyrolysis, particularly catalytic processes, has emerged as an efficient approach for breaking down cellulose and converting it into valuable chemicals. This underscores the importance of optimizing process conditions to achieve improved product yields and efficiencies (Huang et al., 2022).

The oxidative treatment of biochar can significantly enhance its specific surface area, boosting its effectiveness in applications such as adsorption, catalysis, and soil amendment. This oxidation process also promotes the formation of functional groups, including phenolic and carboxylic compounds. Aliyu et al. reported that the initial thermal decomposition phase, primarily involving moisture evaporation, occurs below 150°C, a phenomenon common to lignocellulosic biomass (Abdullah et al., 2017). Pyrolysis typically progresses through three distinct stages—initial dehydration, active decomposition of hemicellulose and cellulose, and slow degradation of lignin—each characterized by specific temperature ranges and gas evolution profiles. These stages are summarized schematically in Fig. 2.

During the second stage, the decomposition of hemicellulose leads to substantial weight loss as weaker bonds in the biomass are replaced with stronger ones, forming biochar and volatile compounds (Abdullah et al., 2017; Bach & Chen, 2017). The resulting biochar has shown potential as a cost-effective alternative to activated carbon for pollutant filtration and carbon sequestration. Additionally, its use in direct fuel applications, nutrient retention enhancement, and soil productivity improvement highlights its multifunctionality in addressing both environmental and agricultural challenges (Sajdak et al., 2023).

3.2 Characteristics of Palm Tree Waste

The composition of palm tree waste was analyzed to quantify its cellulose, hemicellulose, lignin, and elemental constituents. The lignocellulosic fractions were determined using a gravimetric approach, while proximate and ultimate analyses were conducted in accordance with standardized ASTM procedures. These analyses included the determination of ash,

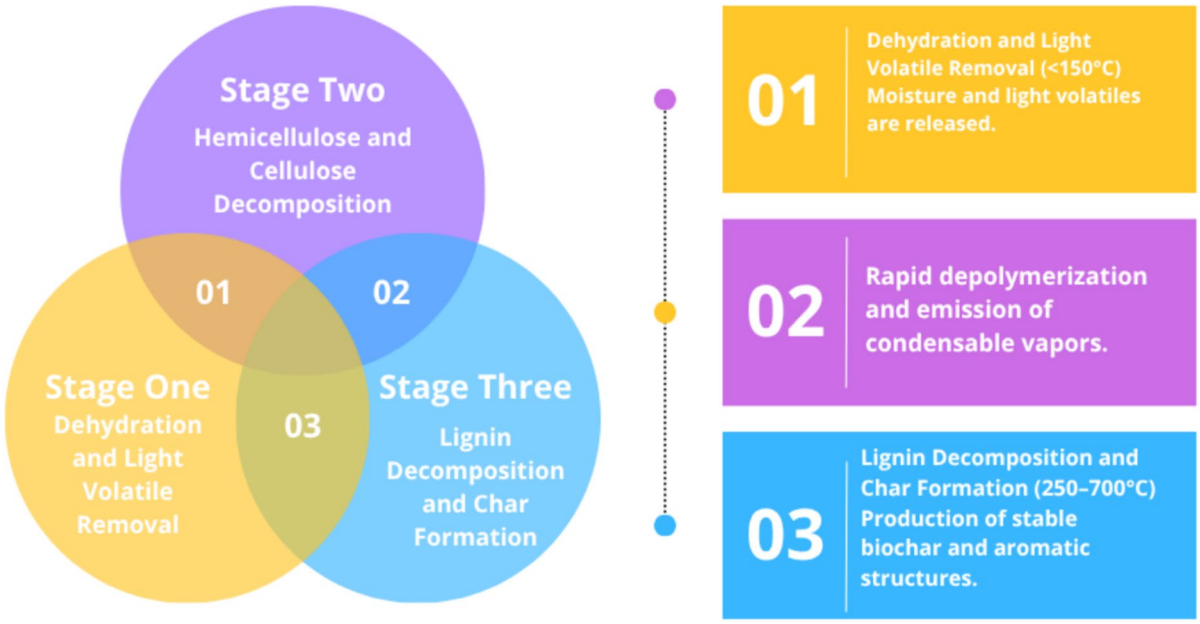


Fig. 2 Schematic illustration of the three main stages of biomass pyrolysis

moisture, and fixed carbon contents, as illustrated in Fig. 3.

The ultimate analysis of carbon, hydrogen, and nitrogen was conducted using a TruSpec CHN analyzer (LECO Corporation). Table 2 provides a comprehensive summary of these results, highlighting the high lignin content of palm tree waste, which contributes to the production of biochar with enhanced structural integrity.

These analyses confirm that palm tree waste, due to its high volatile matter and substantial lignin content, holds great promise for pyrolysis applications. Its lignin structure contributes to the production of biochar with a high molecular weight, while

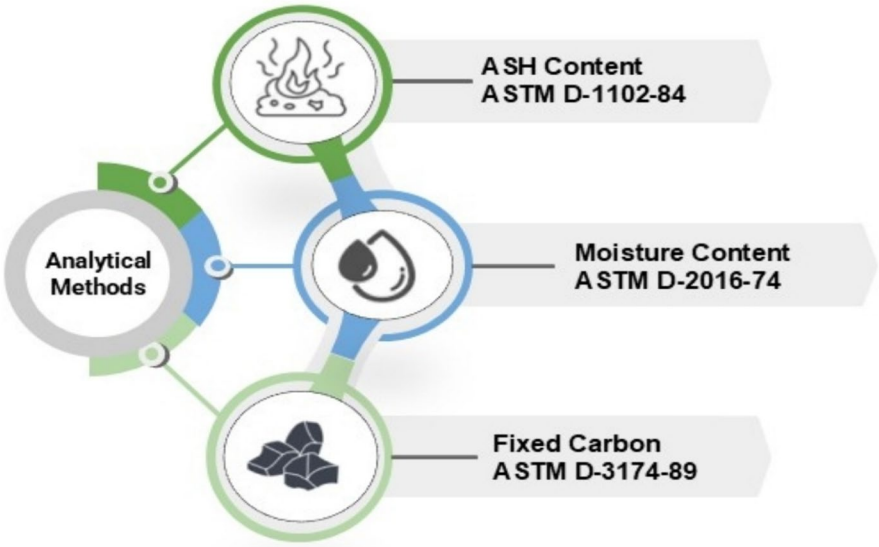


Fig. 3 Schematic representation of the ASTM standard methods used for proximate and ultimate analyses of palm waste. The analysis procedures include determination of ash content (ASTM D-1102–84), moisture content (ASTM D-2016–74), and fixed carbon (ASTM D-3174–89). This illustration summarizes the methodological workflow applied to evaluate the physicochemical properties of palm biomass before pyrolysis

Table 2 Feedstocks' physicochemical characteristics and those of biochar weight percentage

Value	Unit	Parameter	Component
27.7 ± 1.2	(%)	Cellulose	
21.6 ± 0.9	(%)	Hemicellulose	
44.0 ± 1.5	(%)	Lignin	
986 ± 25	(kg/m^3)	Density	**Proximate analysis**
0.15 ± 0.01	(%)	Ash content (%)	
13.6 ± 0.8	(%)	Water content (%)	
26.6 ± 0.5	(MJ/kg)	HHV	
2.5 ± 0.1	(–)	PH	
0.15 ± 0.01	(%)	S	**Ultimate analysis**
26.0 ± 1.0	(%)	O	
8.5 ± 0.4	(%)	H	
85.9 ± 1.3	(%)	C	

All experiment versions in Table 2 were replicated three times, and the estimated standard deviation was based on just laboratory experiments

the reactive cellulose and hemicellulose components facilitate the generation of bio-oil and other volatiles at lower temperatures.

3.2.1 Experimental Setup for Pyrolysis

Pyrolysis experiments were conducted using a precision-engineered reactor system designed to optimize control over critical parameters. A 30 g sample of palm tree waste was introduced into a high-temperature-resistant plastic tube reactor, enclosed within an electric vertical furnace. The furnace was equipped with a digital controller to ensure precise temperature regulation, monitored by an internally positioned K-type thermocouple. To enhance process consistency, the reactor's nitrogen flow was carefully regulated at 2 L/min to maintain an inert atmosphere, eliminating oxidation during pyrolysis. The experiments were conducted in multiple series to investigate the impact of temperature and other operational parameters on pyrolysis yields and product distribution. In the initial series, the effect of temperature was analyzed, with temperatures set at 350, 400, 450, 500, and 550 °C, while keeping the particle size fixed at 1.7–2 mm and the reaction time constant at 60 min. Novel advancements in condensation systems allowed for precise quantification of liquid products through an array of condensers calibrated to 0.5°C intervals. The condensers ensured efficient separation of volatile products, enhancing the accuracy of product characterization. Post-reaction, the solid biochar residue was carefully extracted, weighed, and analyzed.

3.2.2 Unique Contributions

The novelty of this study lies in its focus on the underutilized palm tree waste as a feedstock, specifically leveraging its high lignin content (44%) to produce biochar with superior structural and functional properties. Additionally, the tailored reactor configuration and precise control of pyrolysis conditions enabled a deeper investigation into the relationship between biomass composition and product yields, particularly in optimizing bio-oil production (Hamidzadeh et al., 2023).

3.2.3 Chemical Composition and Thermal Reactivity

Palm tree waste consists predominantly of lignocellulosic materials: cellulose (27.7%), hemicellulose (21.6%), and lignin (44%). The high lignin content contributes to increased biochar yield and a higher molecular weight of bio-oil, attributed to the complex aromatic structure of lignin. In contrast, hemicellulose and cellulose, which are thermally less stable, decompose at lower temperatures (250–350°C), facilitating volatile production. Lignin's decomposition occurs across a wider temperature range (250–700°C), allowing for extended char formation and additional thermal stability (Sait et al., 2012).

Compared to other biomass residues, palm waste demonstrated enhanced pyrolysis efficiency due to its high volatile matter content (70.9%) and favorable elemental composition, including 45.7% carbon and 48.9% oxygen, making it an excellent candidate for bio-oil production (Abnisa et al., 2013).

3.3 Thermal Gravimetric Analysis (TGA)

The thermal behavior and decomposition pathways of palm biomass-derived biochar were analyzed through Thermal Gravimetric Analysis (TGA) under well-controlled experimental conditions. Four distinct samples were studied to investigate the influence of temperature, feed rate, and the presence of an inert atmosphere (nitrogen gas). The detailed experimental setup and operating conditions for each run are summarized in Table 3.

Table 3 Experimental design matrix showing the temperature range, feed rate, and nitrogen gas injection conditions used in the pyrolysis experiments

Experiment ID	Temperature (°C)	Feed rate (g/min)	Nitrogen gas injection	Description
T450F30N1	450	30	Yes	Low-temperature run with nitrogen flow
T500F30N1	500	30	Yes	Medium-temperature run under nitrogen atmosphere
T550F30N1	550	30	Yes	High-temperature run under nitrogen atmosphere
T500F20N0	500	20	No	Medium temperature, low feed rate, without nitrogen flow

The TGA and Differential Thermal Gravimetric (DTG) profiles for these samples revealed distinct thermal decomposition stages:

1. Moisture evaporation: Initial weight loss occurred within a temperature range of 47.23°C to 76.83°C, depending on sample conditions. Samples subjected to nitrogen injection exhibited lower evaporation temperatures due to the increased thermal efficiency provided by the inert atmosphere. This stage accounted for approximately 5% of weight loss.
2. Volatile release: The subsequent phase involved the degradation of hemicellulose and cellulose, starting at approximately 200 °C and peaking at around 400 °C, as indicated by pronounced DTG peaks. This phase marked the breakdown of weak polymeric bonds within the biomass structure, resulting in the formation of volatile organic compounds.
3. Char formation and lignin decomposition: The final stage, extending up to 700 °C, corresponded to the slow decomposition of lignin and the formation of biochar. Lignin's aromatic and cross-linked structure contributed to its thermal stability, yielding significant char residues with robust structural integrity.
4. Ash combustion: At temperatures exceeding 800 °C, the remaining biochar underwent combustion, leaving residual ash.

3.3.1 Key Observations and Novel Insights

Temperature Dependency of Biochar Yield Biochar production peaked at 500 °C, with yields reaching 40% of biomass weight. This observation underscores the suitability of slow pyrolysis conditions for maximizing solid residue production while preserving its structural quality.

Nitrogen Atmosphere Impact Samples processed in a nitrogen environment exhibited enhanced weight loss efficiency and sharper DTG peaks during volatile release, reflecting improved heat transfer and reduced thermal oxidation. Kinetics of Decomposition: The TGA-DTG profiles revealed distinct reaction pathways for hemicellulose, cellulose, and lignin. Hemicellulose, with its amorphous structure, decomposed rapidly at lower temperatures, whereas lignin's aromatic complexity contributed to sustained thermal resistance, enhancing biochar yield and quality.

3.3.2 Significance and Application

The results of the TGA and DTG analyses offer critical insights into the thermal stability and decomposition kinetics of palm tree biomass. This study provides a robust framework for optimizing pyrolysis conditions to tailor biochar and bio-oil yields, aligning with global efforts to develop sustainable energy and carbon sequestration solutions. Furthermore, the findings emphasize the potential of palm biomass as a cost-effective feedstock for high-performance biochar, suitable for applications such as soil amendment, pollutant filtration, and renewable energy production.

3.3.3 Thermal Gravimetric Analysis (TGA) Results and Implications

The thermal decomposition of date palm biomass, analyzed through TGA, highlights distinct

degradation phases influenced by the feed conditions and pyrolysis environment as presented in Fig. 4. Hemicellulose decomposition, occurring in the range of ~ 180–390°C, peaked at 330.8°C for sample T500F20N0, characterized by biochar production at 500 °C without nitrogen. For sample T500F20N1, in the presence of nitrogen, hemicellulose degradation was observed at ~ 350 °C, aligning with the gradual breakdown of lignin (~ 170–835°C).

The second weight loss observed in the TGA curves signifies the degradation of lignin and hemicellulose, particularly in sample T550F20N1, where decomposition peaked at 362.68°C, producing biochar with higher carbon content due to the nitrogen atmosphere. This condition enhanced thermal stability, reduced oxygenated compounds, and maximized fixed carbon yield. The moisture content of leaf stems, ~ 30%, accounted for greater initial weight loss, while the reduced volatile matter in stems resulted in combustion concluding earlier (~ 550 °C) than in leaves and seeds (~ 600 °C). At > 800 °C, residual biomass transitioned to ash, consistent with carbon–oxygen reactions. These results demonstrate the potential of nitrogen-assisted pyrolysis to tailor biochar properties for specific applications, including soil enhancement and carbon sequestration.

The thermal decomposition profiles obtained for date palm biomass reveal critical insights into hemicellulose and lignin degradation under varied pyrolysis conditions. The comparative analysis shows:

- Sample T500F20N0 (500°C, no nitrogen): Peak hemicellulose degradation occurred at 330.8°C, consistent with earlier studies showing decomposition within 290–350°C for non-catalytic environments (Zhou, 2024).
- Sample T500F20N1 (500°C, with nitrogen): Enhanced carbon yield with delayed lignin breakdown (~ 350 °C), aligning with studies on nitrogen's role in suppressing secondary oxidation (Yan, 2023).

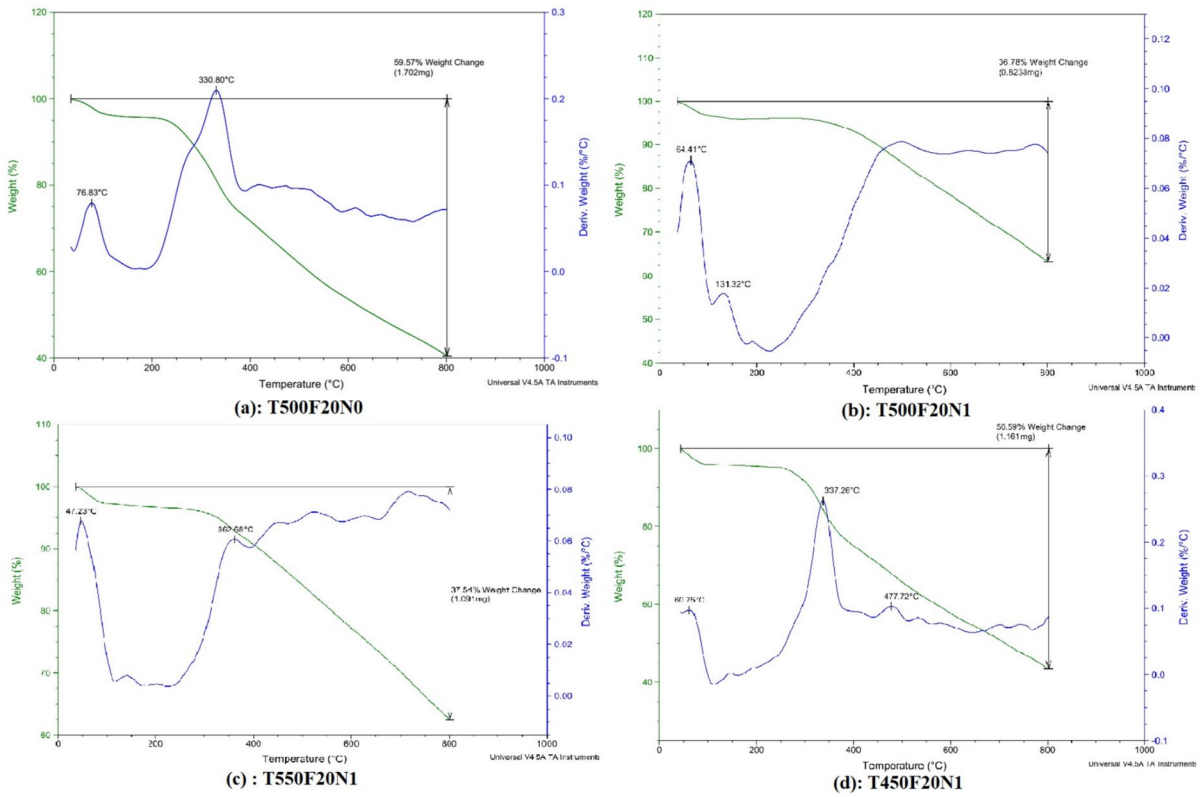


Fig. 4 Results of TGA analysis on different samples

– Sample T550F20N1 (550°C, with nitrogen): Lignin decomposition peaked at 362.68°C, producing biochar with improved thermal stability, similar to optimized biochar in nitrogen-enhanced pyrolysis reported in high-quality journals (Yang, 2007).

This tailored approach emphasizes nitrogen's role in achieving higher fixed carbon content, supported by elemental analysis (C: 45.7%, O: 48.9%). Residual mass stabilization at > 800 °C confirms enhanced char thermal resistance.

3.4 GC–MS (Gas Chromatography-Mass Spectrometer) Analysis of Bio-oil Produced from Palm Tree Waste

According to the GC–MS analysis performed on the samples at 450 °C, more substances were produced, and at higher temperatures, fewer substances of higher quality were produced. The groups of compounds identified during pyrolysis at temperatures of 450 °C, 500 °C, and 550 °C include three main categories: acids, alcohols, and furans. Volatiles produced during pyrolysis were linearly analyzed through GC–MS analysis. The compounds can be sorted into different categories, including acids, ketones, phenols, aldehydes, alcohols, furans, sugars, hydrocarbons, ethers, nitrogen-containing compounds, and sulfur-containing compounds. In Table 4, the chromatogram has the highest peak value related to Melzitus (a type of vegetable sugar) in fast pyrolysis, which was performed at 450 °C in the detection time of 19.92 min. Acids and sugars form the main parts of the volatile substances produced. Hydroxymethylfurfural, which is one of the most important substances seen in this experiment, was detected at minute 12.243. It is one of the most important substances in the production of bioplastics. Also, Pimeclone, which has medicinal use, is found in the compositions of the materials obtained at this degree of pyrolysis at a detection time of 4.979 min with a peak area of 2.65. Another notable compound that can be mentioned is the compound benzoic acid, which was obtained at 11.44 min with a peak area of 10.51. Benzoic acid, a white, crystalline organic compound, is classified within the carboxylic acid family and serves as a crucial starting material for the industrial production of various other organic compounds. In addition to producing methane, biomass can also be used to create electrical and thermal energy and manufacture plastics (e.g., polyesters such as unsaturated polyester resin, polybutylene terephthalate, polyurethane, and elastic fibers) (Lee et al., 2020; Raju et al., 2019).

Straight-chain carboxylic acids such as tetradecanoic, hexadecanoic, and eicosanoic acids are notably present in the sample, comprising a substantial proportion ranging from 6% at 450 °C to 11% at 500 °C, with a slight decrease to 9% at 550°C. Conversely, the same trend was observed for lipid-derived acids, including palmitic acid, oleic acid, lauric acid, and stearic acid. Additional compounds were detected in the volatile flow, including alkyl esters derived from diverse acids, which were acquired at 450 °C and diminished as the temperature rose. Furthermore, various acids were generated from the thermal breakdown of biomass extractives. It is hypothesized that the deacetylation of hemicellulose serves as the primary route for acetic acid production, occurring at lower temperatures and representing one of the initial pyrolytic reactions to yield organic volatiles during fast pyrolysis (Dong et al., 2012). Finally, in the T450F20N1 sample, furans account for about 12% of the peak area, alcoholic functional groups for 33.67%, organic compounds or hydrocarbons for about 12%, acidic compounds for 27%, and sugars, which are the most important substances, for 17% of the peak area. By adding nitrogen instead of acidic compounds, a series of pyridine or indole substances is produced, and aromatic compounds and phenols also increase.

Gas chromatography–mass spectrometry (GC–MS) analysis was conducted to identify the chemical composition of the bio-oil obtained from the pyrolysis of palm oil biomass. The results revealed a diverse mixture of compounds, predominantly phenolic compounds, furans, and cyclopentenones, which are commonly associated with the thermal decomposition of lignocellulosic biomass. Notably, phenol, furfural, and cyclopentenone were among the most abundant constituents, indicating a significant breakdown of lignin and hemicellulose components. These compounds contribute to the energy-rich characteristics of the bio-oil, making it a suitable candidate for biofuel applications. The GC–MS chromatograms confirmed the consistent presence of these major compounds across all experimental runs.

Sugars are the second-most important compound in terms of the amount detected by GC–MS. The

Table 4 Results of GC–MS analysis of bio-oil generated from sample T450F20N1

Detection time (min)	Peak area	Area (%)	Compound	Row
4.18	22,086,251	2.02	Furan, tetrahydro-2,5-dimethoxy	1
4.419	6,979,688	0.64	2,4-Dimethylfuran	2
4.538	8,165,518	0.75	2-Cyclopenten-1-one, 2-methyl	3
4.663	10,136,689	0.93	Ethanone, 1-(2-furanyl)	4
4.808	5,688,001	0.52	Furanone-(5H)2	5
4.979	28,905,150	14.65	Palmitic acid	6
5.337	122,292,449	6.83	Oleic acid	7
5.555	10,709,247	0.98	2-Furancarboxaldehyde, 5-methyl	8
5.628	38,185,655	3.50	2-Furancarboxaldehyde, 5-methyl	9
6.1	113,622,124	10.41	Phenol	10
6.24	22,278,834	2.04	Phenol	11
6.432	19,823,248	1.82	Cyclobutene-1,2-dione, 3,4-dihydroxy-3	12
7.122	41,647,448	3.82	Cyclopentanedione, 3-methyl-1,2	13
7.247	6,506,024	0.60	Cyclopenten-1-one, 2,3-dimethyl-2	14
7.299	20,114,786	1.84	Dodecane, 1,1-dimethoxy-	15
7.62	4,046,239	0.37	Methylamine(Dimethylcyclobutyl-2,2)	16
7.724	37,564,619	3.44	Phenol, 2-methyl-	17
8.269	4,004,442	0.37	p-Cresol	18
8.414	5,745,747	0.53	Mequinol	19
9.156	21,767,799	1.99	Maltol	20
9.691	33,804,899	3.10	Glutaric acid, 2-ethylhexyl 2-methylpentyl ester	21
10.775	21,831,325	2.00	Methyl-2-(2-oxopropyl)furan-3	22
11.444	114,730,538	10.51	Benzoic acid	23
11.626	32,460,696	2.97	Catechol	24
12.243	53,999,617	4.95	Hydroxymethylfurfural-5	25
12.38	8,588,370	0.79	Benzenemethanol, 3-fluoro-	26
13.629	5,070,227	5.46	Methyl Laurate	27
14.158	5,779,176	3.53	Methyl Stearate	28
14.448	40,901,853	3.75	Phenol, 2,6-dimethoxy-	29
15.46	11,737,272	1.08	Bis[aziridyltrimethylamine]−1,1	30
15.37	6,646,824	0.61	Dodecane, 2,5-dimethyl-	31
16.482	8,494,637	0.78	Chrysanthemic acid	32
18.238	5,661,148	0.52	Propanone, 1-(4-hydroxy-3-methoxyphenyl)−2	33
18.833	6,376,969	0.58	beta.-D-Glucopyranose, 1,6-anhydro-	34
19.922	142,244,834	13.03	Melezitose	35
22.693	43,046,634	3.94	Mercaptobenzothiazole-2	36

average proportions of sugar peaks decline as temperatures increase, decreasing from 18.8% at 450 °C to 18% at 550 °C, indicating a greater production of acidic and sugary substances at lower temperatures. Levoglucosan emerges as the primary compound among the released sugars, with its average proportion dropping from about 12% at 450 °C to 1.8% at 500 °C (5% at 550 °C). The second most abundant compound among the sugars displays an average relative concentration varying from 3.6% to 2% over the temperature span of 450–550°C, as illustrated in Table 5. In the palm tree waste samples at 500 °C pyrolysis, aminopyrazine, phenols, and a kind of sugar undergo an increasing process, and the amount of glucose increases from 18.8% at 450 °C to 23.54% at 500°C. Researchers suggested a mechanism to

Table 5 The results of GC–MS analysis of bio-oil generated from sample T550F20N1

Detection time (min)	Peak area	Area (%)	Name of the material	Row
4.175	9,757,661	1.56	Furan, tetrahydro-2,5-dimethoxy-	1
4.492	15,085,245	2.41	Cyclopenten-1-one, 2-methyl-2	2
4.621	7,890,343	5.26	Palmitic acid	3
5.332	140,339,283	22.42	Aminopyrazine 1-oxide-3	4
5.586	39,844,007	2.73	Oleic acid	5
6.038	200,915,928	32.10	Phenol	6
7.044	11,354,570	1.81	Cyclopenten-1-one, 2-hydroxy-3-methyl-2	7
7.262	4,508,991	6.27	Furancarboxaldehyde, 5-methyl-2	8
7.724	7,349,646	1.17	Phenol, 2-methyl-	9
8.347	10,837,230	1.73	p-Cresol	10
9.675	10,167,317	1.62	Glutaric acid, 2-methylpent-3-yl 2-ethylhexyl ester	11
11.268	17,789,429	2.84	1,4:3,6-Dianhydro-.alpha.-d-glucopyranose	12
11.621	19,871,365	3.17	Catechol	13
12.466	5,636,838	0.90	Chlorobenzoic acid, cyclobutyl ester-4	14
13.291	8,478,991	1.35	(Hydroxymethyl)−2-(dimethoxymethyl)furan-5	15
18.916	116,083,315	18.55	beta.-D-Glucopyranose, 1,6-anhydro-	16

explain the breakdown of spruce wood sawdust, highlighting that polymerization into hydrocarbons and furans serves as a primary method for cellulose decomposition. Additionally, they observed that pyranose undergoes ring opening, resulting in the release of volatile oxygen. Consequently, the quantities of sugars and fumes discharged during pyrolysis are comparatively elevated at lower temperatures but diminish as temperatures increase due to cracking reactions (Gu et al., 2013).

3.5 The Results of the Yield of the Products in the Experiments

3.5.1 Formation of Lipid-Derived Acids

The GC–MS analysis confirmed palmitic acid, oleic acid, lauric acid, and stearic acid as the dominant fatty acid compounds from the lipid fraction of palm date residues. Palmitic acid, constituting approximately 50% of the fatty acid methyl esters (FAMEs), was the primary component. Figure 5 illustrates a notable decline in FAMEs, especially palmitic acid, as the pyrolysis temperature increases from 450 °C to 550°C. The reduced mass of palmitic and oleic acids at higher temperatures highlights lipid decomposition during fast pyrolysis. Conversely, at lower temperatures, these acids are predominantly retained in the condensates. This phenomenon is attributed to the partial transesterification of the lipid content, leading to methyl palmitate and methyl oleate formation. This aligns with the temperature-dependent behavior of FAMEs reported in recent studies, emphasizing efficient lipid conversion into bio-oil under controlled thermal conditions.

3.5.2 Determination of Optimal Temperature Conditions of Pyrolysis Products

According to the findings from the Design Expert software modeling (Fig. 6), the increase in temperature positively correlates with both biochar and bio-oil production. However, the optimal temperature for biochar production is identified as 500°C. At this temperature, biochar quality is maximized by preventing excessive thermal degradation into ash, ensuring a high carbon content, and enhancing its porosity, crucial for applications like soil amendment and pollutant adsorption. Similarly, bio-oil production is favored at higher temperatures, with 550 °C yielding the highest bio-oil output.

These results align with the established understanding that pyrolysis temperature is a critical factor influencing both the yield and quality of biochar and bio-oil. Several studies have demonstrated that temperatures in the range of 450–550°C promote efficient

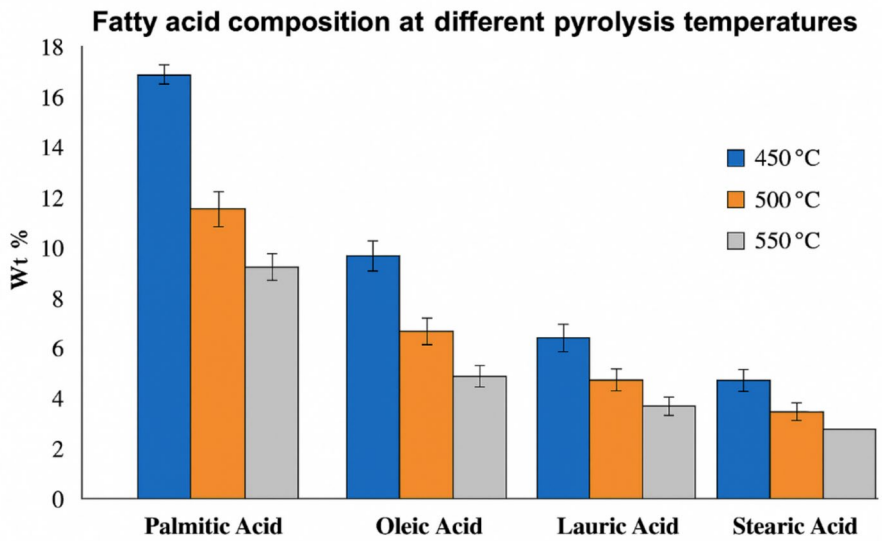


Fig. 5 The generation of FAMEs from palm date residues during fast pyrolysis

biomass conversion while maintaining the desired characteristics of each product. Specifically, 500 °C has been noted as an ideal temperature for producing high-quality biochar, as it balances between the carbon-rich output and minimal ash formation (Li et al., 2016; Sadhukhan, et al., 2014). Moreover, higher temperatures (e.g., 550 °C) tend to favor the production of volatile compounds and bio-oil, although this may come at the cost of biochar yield (Tripathi, et al., 2016).

3.5.3 Refined Analysis for Optimal Feeding Rate Conditions

Based on the evaluation of the data in Fig. 7, the optimal feeding rate for pyrolysis falls within the range of 20 to 30 engine revolutions per minute (g/min). At this range, a lower feeding rate significantly enhances the yield of biochar and bio-oil. The observed increase in yields can be attributed to the extended residence time afforded to the biomass, which facilitates more effective thermal decomposition and energy transfer.

The high volume and low density of the biomass are critical factors in this process. A reduced feeding rate ensures prolonged exposure to consistent heat, promoting the breakdown of complex organic compounds into valuable pyrolysis products. Additionally, slower feeding rates minimize the likelihood of incomplete combustion or unconverted residues, which are often associated with excessive feeding rates. This behavior suggests that controlling the feeding rate is vital for achieving optimal thermal efficiency and maximizing the conversion of biomass into biofuels and biochar. Balancing the rate of material input with the system's heating capacity is essential for ensuring the uniformity of temperature distribution, which directly impacts the quality and quantity of the pyrolysis outputs. This optimized approach aligns with best practices for efficient thermal conversion in bioenergy production systems.

3.5.4 Refined Analysis: Optimal Conditions for Pyrolysis Under Nitrogen Influence

The controlled introduction of nitrogen into the pyrolysis system demonstrated significant effects on the quality and quantity of biochar production. By creating an inert atmosphere and a partial vacuum, nitrogen mitigated oxidative reactions, enhancing the carbonization process and leading to increased biochar yields. The inert nature of nitrogen prevents secondary reactions that could degrade the biochar, contributing to improved structural integrity and thermal resistance.

However, the presence of nitrogen correlated with a slight reduction in the overall biowaste conversion efficiency, possibly due to the displacement of

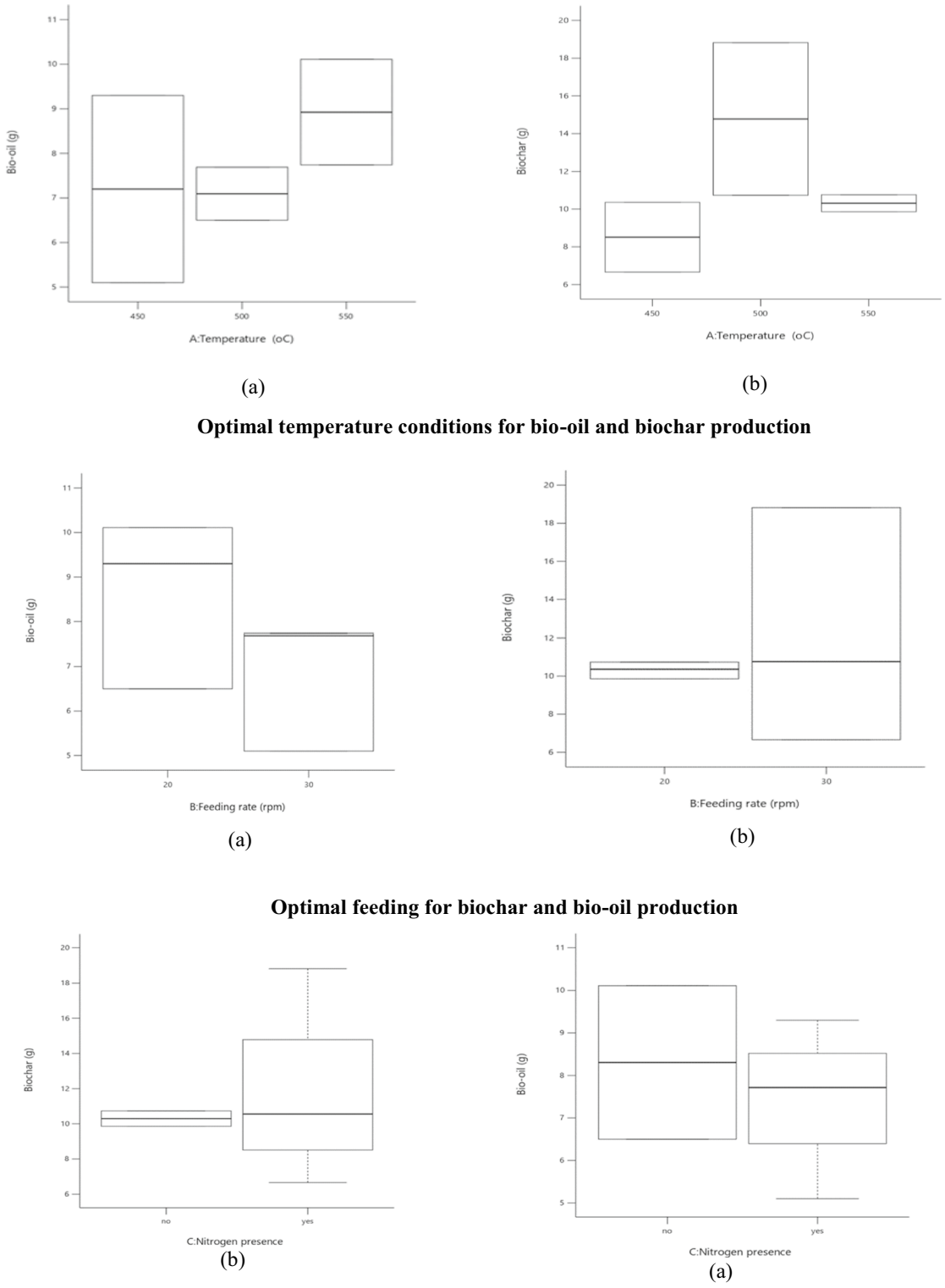


Fig. 6 Optimal conditions of pyrolysis variables: **A.** temperature, **B.** feeding rate, **C.** under influence with and without nitrogen for biochar and bio-oil production

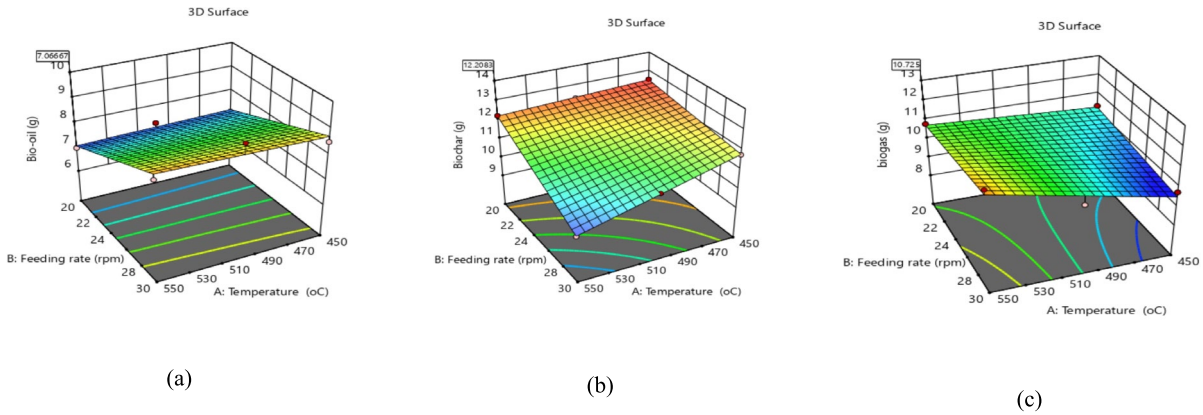


Fig. 7 Optimum conditions for palm pyrolytic products: **a**) bio-oil production, **b**) biochar production, and **c**) biogas production

oxygen, which is necessary for partial combustion and faster decomposition in oxygen-rich environments. The trade-off highlights a critical consideration: while nitrogen enhances the quality of biochar, it slightly lowers the production of bio-oil and biogas due to the suppression of certain volatile organic compound formations. The improved bioavailability and stability of the biochar in nitrogen-enhanced pyrolysis conditions underscore its potential for applications in soil amendment, carbon sequestration, and environmental remediation. Thus, nitrogen's role in optimizing pyrolysis is pivotal, emphasizing the importance of tailoring process conditions based on the desired end products and their applications.

3.6 EDX Analysis (Scanning Electron Microscope Analysis) of Biochar

Energy Dispersive X-ray (EDX) analysis via scanning electron microscopy provided detailed insights into the elemental composition of biochar produced at varying pyrolysis temperatures. Results indicated that biochar generated at higher temperatures exhibited elevated carbon content, correlating with intensified carbonization and thermal decomposition of cellulose, hemicellulose, and lignin in the biomass. This suggests increased thermal resistance of biochar as pyrolysis temperature rises (Sizirici et al., 2021). Additionally, oxygen content decreased with increasing temperature due to the evaporation of oxygen-containing compounds, as exemplified in the T550F20N1 sample, where oxygen content was measured at 21.32%. Higher pyrolysis temperatures also promoted calcium enrichment through calcination, leading to the formation of calcium carbonate and its subsequent conversion to calcium oxide. Similarly, magnesium content increased due to the formation of insoluble periclase (MgO). These compositional changes enhance biochar's structural and chemical stability, making it more suitable for applications such as soil amendment and carbon sequestration (Usman et al., 2015).

Notably, the presence of nitrogen in the pyrolysis environment influenced the elemental balance. At 500 °C, biochar produced without nitrogen showed higher carbon and calcium contents, alongside reduced oxygen levels, compared to samples generated under nitrogen. This highlights the nuanced role of the pyrolysis atmosphere in determining biochar properties (Sizirici et al., 2021). Furthermore, trace elements like aluminum and silicon were detected, attributed to inherent metal oxides in the biomass. The overall reduction in oxygen, hydrogen, and nitrogen content with temperature increase aligns with the loss of surface functional groups, driven by evaporation and devolatilization processes. Figure 8 illustrates these trends, emphasizing the temperature-dependent chemical transformations that shape biochar composition and functionality (Usman et al., 2015).

The pyrolysis process enriches biochar with mineral-based components, notably calcium, magnesium, potassium, and sodium, due to the thermal decomposition of biomass and subsequent concentration of inorganic constituents in the residual matrix. The highest total basic cation content in biochar is

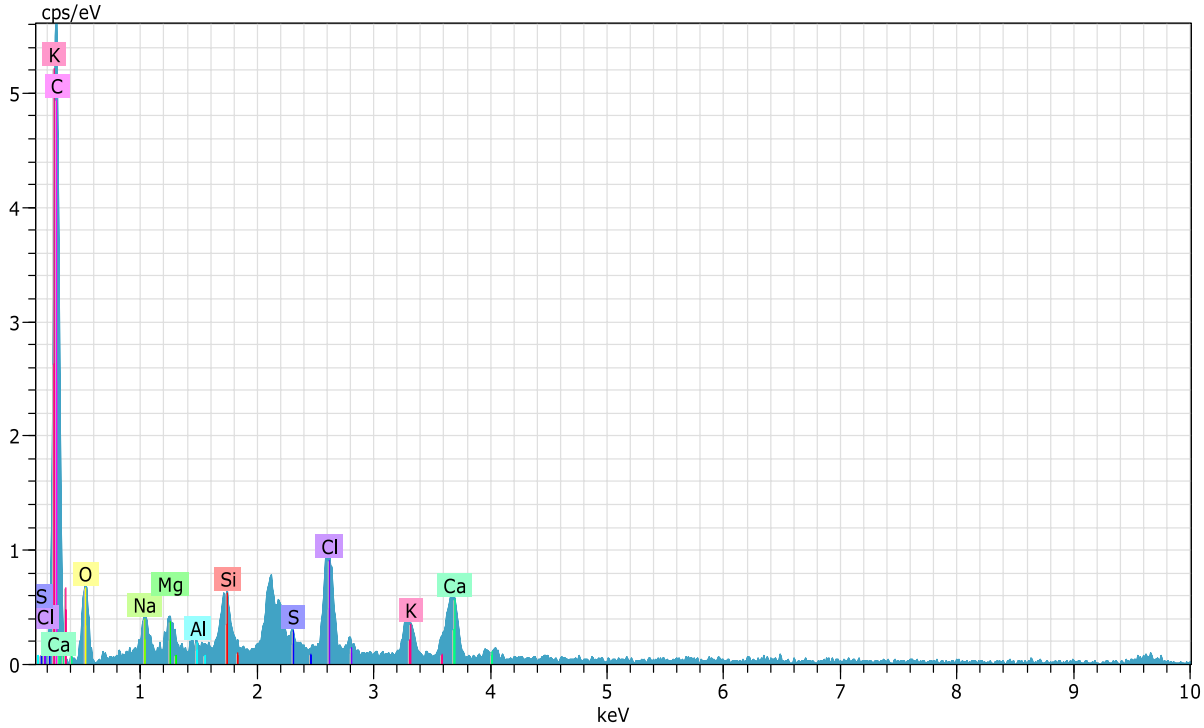


Fig. 8 EDX analysis of sample; T450F20N1

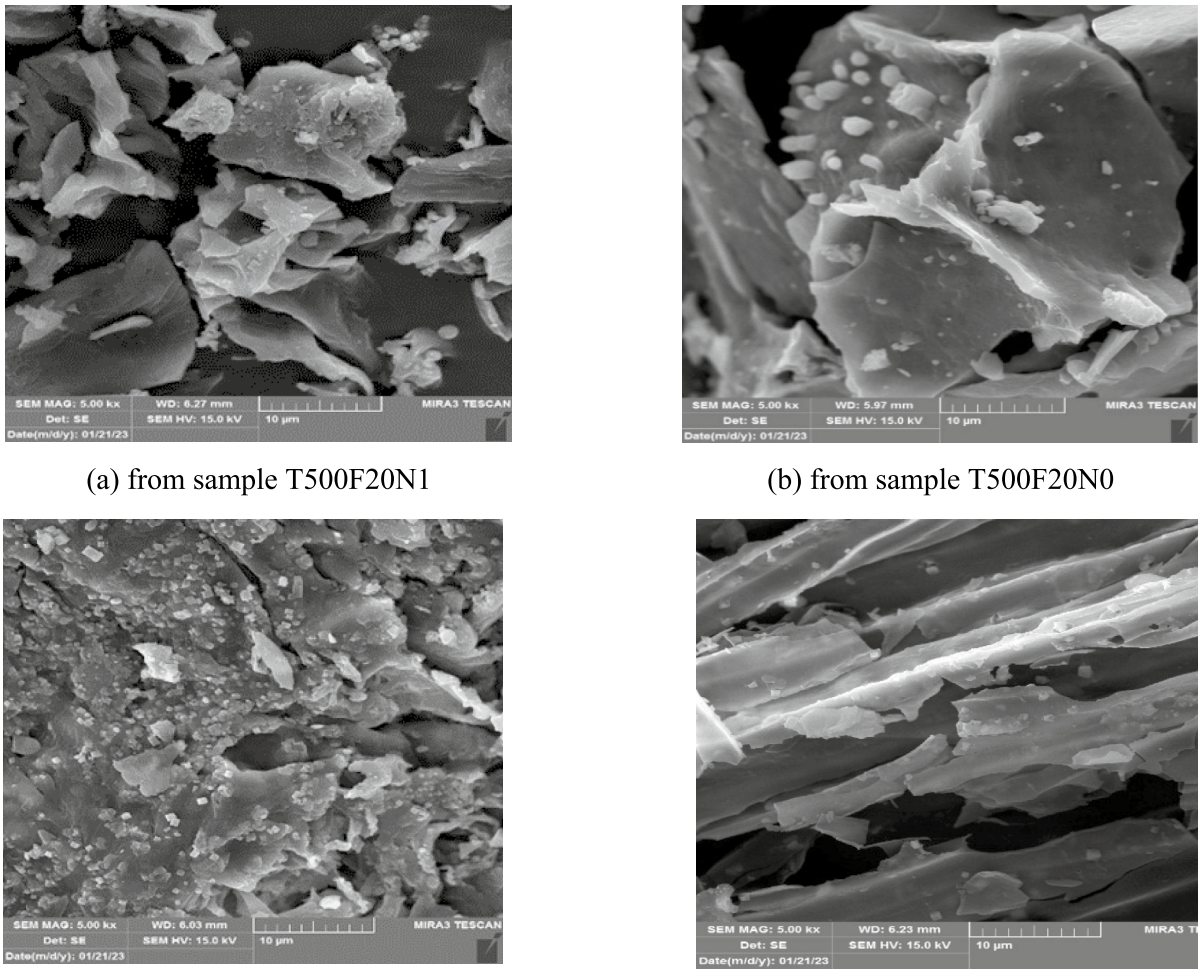
observed at elevated pyrolysis temperatures, such as 800 °C, primarily due to mineral enrichment and increased ash content. This enrichment, while advantageous for specific applications like soil amendment, may also influence biochar's porosity and specific surface properties (Ahmad et al., 2012).

However, excessive ash content can potentially reduce the biochar's porosity and adsorption efficiency, which are critical in applications like water purification or pollutant capture, including H/C and O/C, and serve as valuable indicators of biochar's chemical stability and polarity. For example, biochar with higher H/C ratios, typically produced at lower pyrolysis temperatures, reflects incomplete thermal decomposition, which limits its long-term stability for carbon sequestration. Conversely, biochar derived from date palm residues at 500 °C, characterized by a volatile matter content below 10%, demonstrates substantial potential for carbon sequestration due to its lower oxygen content, increased aromaticity, and enhanced resistance to biodegradation (Zhao et al., 2018). At higher pyr processes, such as calcination, contribute to the transformation of calcium and magnesium into their oxide forms, like calcium oxide and magnesium oxide. These oxides enhance biochar's reactivity and buffering capacity, which are advantageous in agricultural applications and soil remediation. The balance between thermal stability, enriched cation content, and functional properties at moderate temperatures underscores the versatility of biochar for multiple applications, ranging from soil amendment to carbon capture and storage.

3.7 SEM Analysis of Biochar from Palm Date Residues

The scanning electron microscopy (SEM) analysis of biochar samples derived from palm date residues revealed significant changes in surface morphology with increasing pyrolysis temperature, as shown in Fig. 9. The quantity and size of pores on the biochar's surface expanded with higher thermal decomposition temperatures, indicating enhanced porosity.

Comparative SEM micrographs demonstrated that porous structures, resulting from the evaporation of organic matter, became more pronounced



(a) from sample T500F20N1

(b) from sample T500F20N0

(c) from sample T450F20N1

(d) from sample T550F20N1

Fig. 9 Morphological analysis of biochar before and after treatment

as pyrolysis temperature increased. At higher temperatures, the formation of channels and deeper pores was evident. These observations suggest that the pyrolysis process preserved the fibrous cell structures of the palm skin and mesocarp, despite chemical decomposition leading to water and organic component loss. This structural retention highlights the incomplete disintegration of the palm biomass, reducing its overall mass. Biochar particles formed at lower temperatures exhibited smoother surfaces and less fragmentation compared to those produced at 500 °C and 550°C. Particles generally displayed irregular shapes, with surface textures varying from smooth to rough. This variation is attributed to the behavior of lignin, which undergoes thermal softening and decomposition differently depending on the temperature. Interestingly, SEM micrographs indicated that the pyrolysis process did not create new pores beyond those originally present in the biomass skeleton. Additionally, the impact of pyrolysis temperature on microscopic appearance was minimal. These findings align with nitrogen porosimetry results, which showed limited porosity development across the tested temperature range, suggesting that intrinsic factors within the biomass largely govern the final porosity characteristics (Dowson & Styring, 2017).

3.8 Mechanism of Value-Added Product Formation from Palm Date Residues

The mechanism for the formation of fatty acid methyl esters (FAMEs) from palm date residues during pyrolysis is a proposed pathway depicted in Fig. 10. The process suggests a complex interaction between triglycerides and lignin, the key components of the biomass. Triglycerides, serving as the primary lipid source, undergo thermal decomposition, releasing fatty acids, which subsequently undergo methylation. In this mechanism, the fatty acid moieties are methylated using a methyl group donor, with the methoxy groups present in the phenolic units of lignin being a plausible candidate (Harth et al., 2023; Trajano et al., 2013).

This interaction could indicate a synergistic relationship between triglycerides and lignin during pyrolysis. Based on this hypothesis, the methoxy groups in lignin could potentially react with the carboxyl groups of fatty acids, resulting in the production of methyl palmitate and methyl oleate. The identification of these compounds is consistent with, and strongly supports, the proposed interplay between lipid content and lignin units in FAME formation. It is important to note that while other methylation pathways may exist, the role of lignin-derived methoxy groups is considered highly plausible given the established reactivity of these functional groups under thermal conditions (Harth et al., 2023; Trajano et al., 2013). This role becomes particularly evident at lower pyrolysis temperatures, where conditions appear to favor this specific methylation pathway over the extensive decomposition of lignin. Moreover, the study observed that increasing pyrolysis temperature shifted the product distribution towards aromatic compounds such as benzene, guaiacol, and syringol, further underscoring the temperature-dependent competition between the stabilization of fatty acids as FAMEs and the fragmentation of lignin. This duality in the reaction pathways underscores the importance of optimizing pyrolysis conditions to balance the production of diverse value-added products. The proposed mechanism provides a valuable framework for understanding the intricate chemistry of biomass pyrolysis, highlighting the potential contribution of both triglycerides and lignin in determining the yield and composition of pyrolysis products. This understanding can guide the development of tailored strategies to enhance the efficiency and selectivity of bio-based chemical production from agricultural residues like palm date biomass.

4 Conclusion

This study successfully demonstrated the thermal conversion of palm date residues into

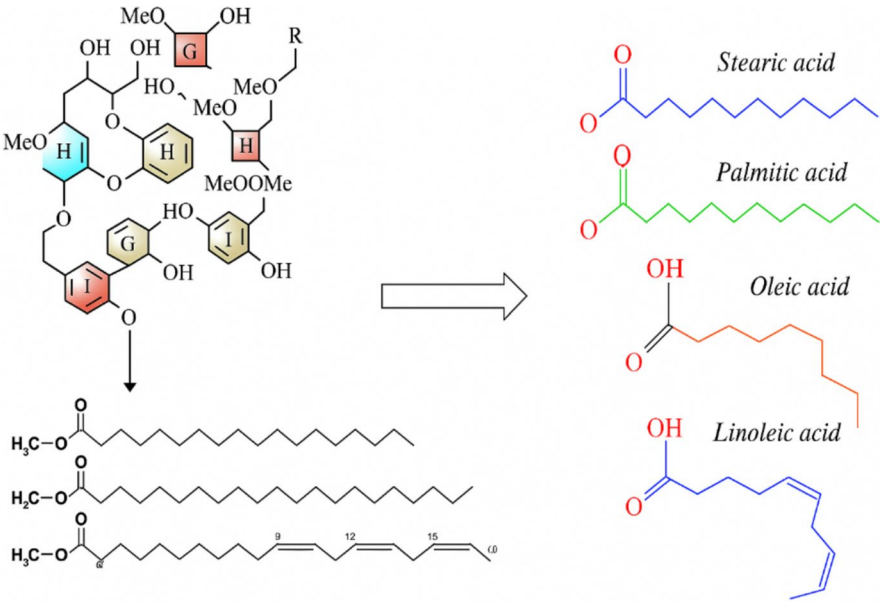


Fig. 10 The mechanism of formation of FAMEs during pyrolysis of palm date residues

valuable biochar and fatty acid methyl esters (FAMEs) through fast pyrolysis in a fluidized bed reactor. The findings provide a comprehensive understanding of how key operational parameters—temperature, feeding rate, and nitrogen atmosphere—govern product yield, composition, and physicochemical characteristics. The optimization of these variables revealed that introducing nitrogen during pyrolysis not only enhanced FAMEs yield but also improved the structural stability and thermal resistance of the resulting biochar. Conversely, higher temperatures favored aromatic compound formation, suggesting the potential for tailoring process conditions to obtain specific bio-based products for industrial applications.

The mechanistic insights obtained from GC–MS, FESEM, and TGA analyses clarified the molecular interactions responsible for FAME formation, particularly the methylation of triglycerides through methoxy groups originating from lignin degradation. This synergistic pathway represents a novel contribution to understanding lipid–lignin co-reactions in biomass pyrolysis and can guide future catalyst design and process integration strategies for enhanced biofuel generation. Beyond technical findings, the study offers significant environmental and economic implications. By valorizing abundant agricultural residues, particularly in regions like Iran where palm waste disposal remains a major challenge, this research presents a sustainable alternative to open burning, effectively reducing greenhouse gas emissions and promoting circular bioeconomy principles. The production of both FAMEs and functional biochar underscores the feasibility of achieving dual benefits—renewable fuel generation and soil-enhancing carbon materials—within a single thermochemical platform.

Overall, the study contributes to advancing sustainable waste-to-energy technologies by elucidating key thermochemical pathways, identifying optimal reaction conditions, and demonstrating the dual utility of palm biomass. Future research should focus on pilot-scale implementation, catalyst integration to further enhance selectivity, and comparative life-cycle assessments to quantify the long-term environmental benefits. This work thus provides a strong scientific basis for scaling up pyrolysis-based valorization of agricultural residues toward industrial adoption and sustainable energy systems.

Author Contributions Conceptualization: B.Y., P.G.; Data curation: Z.H., B.Y.; Investigation: Z.H., B.Y.; Methodology: B.Y., P.G.; Software: Z.H.; Visualization: Z.H.; Writing – Original Draft: Z.H., B.Y., P.G.; Writing—Review & Editing: Z.H., B.Y., P.G.; Supervision: B.Y.

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Data Availability Data used in the current study will be available upon request.

Declarations

Ethics Approval Not applicable.

Consent to Participate All the authors listed have participated in this work and have seen the manuscript.

Consent for Publication All the authors listed have agreed to submit it to your journal.

Competing Interest The authors have no relevant financial or non-financial interests to disclose.

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